



SPIDER VAULT 120.0 — BESS

🔌 Usage	Large Factories + Data Centres + Hospitals + Full Campus Power
📡 Output Signal	Pure Sine Wave — THD < 3% — Zero voltage distortion
⚡ Surge Capacity	Up to 180 kW (1.5× rated) for 10 s — Lifts, CNC, HVAC motors
₹ Running Cost	₹0/hr via Solar · 156 kW solar input covers full load continuously
🕒 Transfer Time	10 ms — Zero interruption — Servers, MRI machines stay live
🔊 Noise Level	< 50 dB — Indoor installation, no soundproof room needed
🌿 Emissions	Zero Emissions — No CO ₂ , no NOx — Green Building compliant
🔋 Backup & Solar	120 kWh Battery + 156 kW Solar (10 MPPT) — Continuous operation
🕒 Life / Warranty	Up to 12 Years · 5 Yr Warranty + Extended 12 Yr Option
🛠 Maintenance	Zero Maintenance — No oil, no filters, no AMC contract needed
📶 Connectivity	Ethernet · RS485 · CAN · Remote Monitoring
📈 System Efficiency	97.6% Max Efficiency · >99% MPPT Efficiency

VS

125 KVA DIESEL GENERATOR SET

🔌 General 3-phase loads only — not suitable for critical IT/medical loads	×
📡 Alternator output — high THD damages VFDs, UPS, medical equipment	×
⚡ Fixed rated output — trips on inrush currents from motors & lifts	×
₹ ₹2,025/hr diesel — 22.5 L/hr at full load @₹90/L (Cummins C125D5P)	×
🕒 10–30 sec startup delay — data loss, production line stoppages	×
🔊 75–85 dBA — Dedicated acoustic enclosure + exhaust stack required	×
☁ CO ₂ + NOx + particulates — CPCB IV+ permits, exhaust ducting needed	×
🛢 ~5–6 hrs (120 L tank) — bulk diesel storage, fire safety regulations apply	×
🕒 5–7 Year lifespan · 1–2 Year Warranty (Cummins / Kirloskar)	×
🛠 Oil change every 200–500 hrs · AMC ₹45,000–₹65,000/yr + spares	×
📶 Basic controller only — no cloud, no remote diagnostics, no SCADA	×
📈 ~30–35% thermodynamic efficiency — 65–70% energy lost as heat & exhaust	×

